Notes on Uniform Law of Large Numbers

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August 10, 2026

1 Empirical Process Theory

1.1 Abstract Glivenko-Cantelli Theorem (Theorem 19.4 of Van der Vaart (1998) or Theorem 2.4.1 of Van der Vaart & Wellner (1996))

Every class $\mathcal{F}$ of measurable functions such that $N_{[]}(\varepsilon, \mathcal{F}, L_1(P)) < \infty$ for every $\varepsilon > 0$ is P-Glivenko-Cantelli i.e., $||\mathbb{P}_n - P||_{\mathcal{F}} := sup_{f\in\mathcal{F}}|\mathbb{P}_n f - Pf| \to_{as} 0$.

Proof:

Given a $\varepsilon > 0$, as $N_{[]}(\varepsilon, \mathcal{F}, L_1(P)) < \infty$ by hypothesis i.e., the minimum number of ε-brackets with $L_1(P)$ norm needed to cover $\mathcal{F}$ is finite, there is a finite sequence of ε-brackets with $L_1(P)$ norm such that $\mathcal{F} \subseteq \cup_i[l_i, u_i]$ where $||u_i - l_i||_{P,1} := (P|u_i - l_i|^1)^{1/1} = P(u_i - l_i) < \varepsilon$. For any $f \in \mathcal{F}$ and any i in the finite sequence,

$$(\mathbb{P}_n - P)f = \mathbb{P}_n f - Pu_i + Pu_i - Pf \leq (\mathbb{P}_n - P)u_i + P(u_i - f) \leq (\mathbb{P}_n - P)u_i + P(u_i - l_i) < (\mathbb{P}_n - P)u_i + \varepsilon$$

since $l_i \leq f \leq u_i$. Thus,

$$sup_{f\in\mathcal{F}}(\mathbb{P}_n - P)f < max_j(\mathbb{P}_n - P)u_j + \varepsilon$$

Given x, $max_j(\mathbb{P}_n - P)u_j(x) \to_{as} 0$ as $n \to \infty$ by strong law of large numbers for a sequence of iid random variables. Similarly,

$$(\mathbb{P}_n - P)f = \mathbb{P}_n f - Pl_i + Pl_i - Pf \geq (\mathbb{P}_n - P)l_i + P(l_i - f) \geq (\mathbb{P}_n - P)l_i + P(l_i - u_i) > (\mathbb{P}_n - P)u_i - \varepsilon$$

Thus,

$$inf_{f\in\mathcal{F}}(\mathbb{P}_n - P)f > min_j(\mathbb{P}_n - P)u_j - \varepsilon$$

Given x, $min_j(\mathbb{P}_n - P)u_j(x) \to_{as} 0$ as $n \to \infty$ by strong law of large numbers for a sequence of iid random variables. Together,

$$-\varepsilon < lim_{n\to\infty}inf_{f\in\mathcal{F}}(\mathbb{P}_n - P)f \leq lim_{n\to\infty}sup_{f\in\mathcal{F}}(\mathbb{P}_n - P)f < \varepsilon \qquad a.s.$$

Therefore, $|lim_{n\to\infty}sup_{f\in\mathcal{F}}(\mathbb{P}_n - P)f| = lim_{n\to\infty}sup_{f\in\mathcal{F}}|(\mathbb{P}_n - P)f| := lim_{n\to\infty}||\mathbb{P}_n - P||_{\mathcal{F}} < \varepsilon$ a.s. Q.E.D.

1.2 Corollary 1 (Pointwise Compact Class) (Example 19.8 of Van der Vaart (1998))

If $\mathcal{F} = \{f_\theta : \theta \in \Theta\}$ where (i) f_θ is a measurable and continuous function of θ. (ii) The envelope (i.e., dominating) function of f is integrable. (iii) Θ is a compact set. Then, $N_{[]}(\varepsilon, \mathcal{F}, L_1(P)) < \infty$ (and $sup_{f\in\mathcal{F}}|\mathbb{P}_n f - Pf| \to_{as} 0$ by Abstract Glivenko-Cantelli Theorem)

Proof:

Let $B_i \subset \Theta$ with $center(B_i)$ $\theta_i \in \Theta$ and B_i is an open set. We construct the bracket $[f_{B_i}, f^{B_i}]$ where $f_{B_i} := inf_{\theta \in B_i} f(\theta)$ and $f^{B_i} := sup_{\theta \in B_i} f(\theta)$,

$$\begin{aligned}
||f_{B_i}, f^{B_i}||_{P,1} &= (P|f^{B_i} - f_{B_i}|^1)^{1/1} \\
&= P(f^{B_i} - f_{B_i}) \\
&= Pf^{B_i} - Pf_{B_i} \\
&= \int f^{B_i} dP - \int f_{B_i} dP
\end{aligned}$$

Let $\{B_{im}\}_{m=1}^{\infty}$ be the sequence of B_i with the same center θ_i and the radius of them converges to zero as $m \to \infty$. Thus,

$$\begin{aligned}
lim_{m\to\infty} \int f^{B_{im}} dP - lim_{m\to\infty} \int f_{B_{im}} dP &= \int lim_{m\to\infty} f^{B_{im}} dP - \int lim_{m\to\infty} f_{B_{im}} dP \\
&\qquad \text{by Dominated Convergence Theorem with (ii)} \\
&= \int f^{lim_{m\to\infty} B_{im}} dP - \int f_{lim_{m\to\infty} B_{im}} dP \qquad \text{by (i)} \\
&= \int f_{\theta_i} dP - \int f_{\theta_i} dP = 0
\end{aligned}$$

Therefore, the bracket $[f_{B_i}, f^{B_i}]$ has $L_1(P)$-size ε for any $\varepsilon > 0$. As θ_i is arbitrary from Θ, we have $\{[f_{B_i}, f^{B_i}]\}_{i=1}^{\infty}$. Its corresponding $\{B_i\}_{i=1}^{\infty}$ is a cover of Θ because $\cup_{i=1}^{\infty} B_i = \Theta$ where $B_i \subset \Theta$. By hypothesis (iii), $\{B_i\}_{i=1}^{\infty}$ has a finite subcover that also covers Θ i.e., we have $\{B_j\}_{j=1}^{p} \subset \{B_i\}_{i=1}^{\infty}$ such that $\cup_{j=1}^{p} B_j = \Theta$ where $p < \infty$. This implies that the minimum number of ε-brackets with $L_1(P)$ norm needed to cover $\mathcal{F}$ is finite i.e., $N_{[]}(\varepsilon, \mathcal{F}, L_1(P)) < \infty$ Q.E.D.

Remark 1: The consistency of M-estimator e.g., Theorem 5.7 of Van der Vaart (1998) usually (not a must) requires the uniform convergence $sup_{f\in\mathcal{F}}|\mathbb{P}_n f - Pf| \to_{as} = sup_{\theta\in\Theta}|\mathbb{P}_n m_\theta - Pm_\theta| \to_{as} 0$. This example provides the sufficient conditions.

Remark 2: This form of uniform law of large numbers can be derived directly without using the Abstract Glivenko-Cantelli Theorem.

1.3 Uniform Law of Large Numbers (Direct Derivation, Theorem 4.2.1 of Amemiya (1985))

Let $g(y; \boldsymbol{\theta})$ be a measurable function of y for each $\boldsymbol{\theta} \in \Theta$, which is a compact subset of $\mathbb{R}^K$, and a continuous function of $\boldsymbol{\theta} \in \Theta$ for each y. Assume $\mathbb{E}[g(y; \boldsymbol{\theta})] = 0$. Let $\{y_i\}_{i=1}^{\infty}$ be a sequence of i.i.d. random variables such that $\mathbb{E}[sup_{\boldsymbol{\theta}\in\Theta}|g(y; \boldsymbol{\theta})|] < \infty$ (the dominance condition). Then, $sup_{\boldsymbol{\theta}\in\Theta}|N^{-1}\sum_{i=1}^{N} g(y_i; \boldsymbol{\theta})| \to_p 0$ as $N \to \infty$.

Proof:

We have $\{\Theta_j^M\}_{j=1}^{M}$ where $\cup_{j=1}^{M} \Theta_j^M = \Theta$ and $\Theta_j^M \cap \Theta_k^M = \emptyset$ for any $j \neq k$ and $d(\Theta_j^M, \Theta_k^M) \to 0$ as $M \to \infty$. Let $\boldsymbol{\theta}_j \in \Theta_j^M$, $j = 1, \cdots, M$. Define $g_i(\boldsymbol{\theta}) = g(y_i; \boldsymbol{\theta})$ for convenience. For any $\varepsilon > 0$,

$$sup_{\boldsymbol{\theta}\in\Theta}|N^{-1}\sum_{i=1}^{N} g_i(\boldsymbol{\theta})| > \varepsilon \implies \cup_{j=1}^{M}\{sup_{\boldsymbol{\theta}\in\Theta_j^M}|N^{-1}\sum_{i=1}^{N} g_i(\boldsymbol{\theta})| > \varepsilon\}$$

$$
\begin{aligned}
Pr(sup_{\boldsymbol{\theta}\in\Theta}|N^{-1}\sum_{i=1}^{N} g_i(\boldsymbol{\theta})| > \varepsilon) &\le Pr(\cup_{j=1}^{M}\{sup_{\boldsymbol{\theta}\in\Theta_j^M}|N^{-1}\sum_{i=1}^{N} g_i(\boldsymbol{\theta})| > \varepsilon\}) \\
&\le \sum_{j=1}^{M} Pr(sup_{\boldsymbol{\theta}\in\Theta_j^M}|N^{-1}\sum_{i=1}^{N} g_i(\boldsymbol{\theta})| > \varepsilon) \qquad \text{as } Pr(A\cup B) = Pr(A)+Pr(B)-Pr(A\cap B) \\
&= \sum_{j=1}^{M} Pr(sup_{\boldsymbol{\theta}\in\Theta_j^M}|N^{-1}\sum_{i=1}^{N} g_i(\boldsymbol{\theta}) + N^{-1}\sum_{i=1}^{N} g_i(\boldsymbol{\theta}_j) - N^{-1}\sum_{i=1}^{N} g_i(\boldsymbol{\theta}_j)| > \varepsilon) \\
&= \sum_{j=1}^{M} Pr(sup_{\boldsymbol{\theta}\in\Theta_j^M}|N^{-1}\sum_{i=1}^{N} g_i(\boldsymbol{\theta}_j) + N^{-1}\sum_{i=1}^{N}(g_i(\boldsymbol{\theta}) - g_i(\boldsymbol{\theta}_j))| > \varepsilon)
\end{aligned}
$$

As $|a+b| > \varepsilon \implies \{|a| > \varepsilon/2\} \cup \{|b| > \varepsilon/2\}, Pr(|a+b| > \varepsilon) \le Pr(\{|a| > \varepsilon/2\} \cup \{|b| > \varepsilon/2\}) \le Pr(|a| > \varepsilon/2) + Pr(|b| > \varepsilon/2)$

$$
\begin{aligned}
&\le \sum_{j=1}^{M}\{Pr(sup_{\boldsymbol{\theta}\in\Theta_j^M}|N^{-1}\sum_{i=1}^{N} g_i(\boldsymbol{\theta}_j)| > \varepsilon/2) + Pr(sup_{\boldsymbol{\theta}\in\Theta_j^M}|N^{-1}\sum_{i=1}^{N}(g_i(\boldsymbol{\theta}) - g_i(\boldsymbol{\theta}_j))| > \varepsilon/2)\} \\
&= \sum_{j=1}^{M} Pr(|N^{-1}\sum_{i=1}^{N} g_i(\boldsymbol{\theta}_j)| > \varepsilon/2) + \sum_{j=1}^{M} Pr(sup_{\boldsymbol{\theta}\in\Theta_j^M}|N^{-1}\sum_{i=1}^{N}(g_i(\boldsymbol{\theta}) - g_i(\boldsymbol{\theta}_j))| > \varepsilon/2) \\
&= \sum_{j=1}^{M} Pr(|N^{-1}\sum_{i=1}^{N} g_i(\boldsymbol{\theta}_j)| > \varepsilon/2) + \sum_{j=1}^{M} Pr(N^{-1}\sum_{i=1}^{N} sup_{\boldsymbol{\theta}\in\Theta_j^M}|g_i(\boldsymbol{\theta}) - g_i(\boldsymbol{\theta}_j)| > \varepsilon/2) \\
&\qquad \text{by triangle inequality}
\end{aligned}
$$

Now consider applying law of large numbers to $N^{-1}\sum_{i=1}^{N} sup_{\boldsymbol{\theta}\in\Theta_j^M}|g_i(\boldsymbol{\theta}) - g_i(\boldsymbol{\theta}_j)| \to_p \mathbb{E}[sup_{\boldsymbol{\theta}\in\Theta_j^M}|g_i(\boldsymbol{\theta}) - g_i(\boldsymbol{\theta}_j)|]$ as $N \to \infty$,

$$
\begin{aligned}
sup_{\boldsymbol{\theta}\in\Theta_j^M}|g_i(\boldsymbol{\theta}) - g_i(\boldsymbol{\theta}_j)| &\le sup_{\boldsymbol{\theta}\in\Theta_j^M}|g_i(\boldsymbol{\theta})| + sup_{\boldsymbol{\theta}\in\Theta_j^M}|g_i(\boldsymbol{\theta}_j)| && \text{by triangle inequality} \\
&= sup_{\boldsymbol{\theta}\in\Theta_j^M}|g_i(\boldsymbol{\theta})| + |g_i(\boldsymbol{\theta}_j)| \\
&\le sup_{\boldsymbol{\theta}\in\Theta_j^M}|g_i(\boldsymbol{\theta})| + sup_{\boldsymbol{\theta}\in\Theta_j^M}|g_i(\boldsymbol{\theta})| && \text{as } \boldsymbol{\theta}_j \in \Theta_j^M \\
&= 2sup_{\boldsymbol{\theta}\in\Theta_j^M}|g_i(\boldsymbol{\theta})| \\
&\le 2sup_{\boldsymbol{\theta}\in\Theta}|g_i(\boldsymbol{\theta})|
\end{aligned}
$$

The dominance condition hypothesized implies $sup_{\boldsymbol{\theta}\in\Theta}|g_i(\boldsymbol{\theta})|$ is integrable. Thus,

$$
\begin{aligned}
lim_{M\to\infty}\mathbb{E}[sup_{\boldsymbol{\theta}\in\Theta_j^M}|g_i(\boldsymbol{\theta}) - g_i(\boldsymbol{\theta}_j)|] &= \mathbb{E}[lim_{M\to\infty}sup_{\boldsymbol{\theta}\in\Theta_j^M}|g_i(\boldsymbol{\theta}) - g_i(\boldsymbol{\theta}_j)|] && \text{by Dominated Convergence Theorem} \\
&= \mathbb{E}[0] && \text{by continuity of } g \\
&= 0
\end{aligned}
$$

uniformly for j. Picking a large enough M such that $\mathbb{E}[sup_{\boldsymbol{\theta}\in\Theta_j^M}|g_i(\boldsymbol{\theta}) - g_i(\boldsymbol{\theta}_j)|] \le \varepsilon/2$, $Pr(N^{-1}\sum_{i=1}^{N} sup_{\boldsymbol{\theta}\in\Theta_j^M}|g_i(\boldsymbol{\theta}) - g_i(\boldsymbol{\theta}_j)| > \varepsilon/2) \to 0$ as $N \to \infty$.

Now consider applying law of large numbers to $N^{-1}\sum_{i=1}^{N} g_i(\boldsymbol{\theta}_j) \to_p \mathbb{E}[g_i(\boldsymbol{\theta}_j)] = 0$ as $N \to \infty$. This implies $Pr(|N^{-1}\sum_{i=1}^{N} g_i(\boldsymbol{\theta}_j)| > \varepsilon/2) \to 0$ as $N \to \infty$. Together, $Pr(sup_{\boldsymbol{\theta}\in\Theta}|N^{-1}\sum_{i=1}^{N} g_i(\boldsymbol{\theta})| > \varepsilon) = Pr(|sup_{\boldsymbol{\theta}\in\Theta}|N^{-1}\sum_{i=1}^{N} g_i(\boldsymbol{\theta})| - 0| > \varepsilon) \to 0$ as $N \to \infty$. i.e., $sup_{\boldsymbol{\theta}\in\Theta}|N^{-1}\sum_{i=1}^{N} g_i(\boldsymbol{\theta})| \to_p 0$ as $N \to \infty$. Q.E.D.

Remark 1: The assumption $\mathbb{E}[g(y;\boldsymbol{\theta})] = 0$ is made without loss of generality. It is because we can replace $g(y;\boldsymbol{\theta})$ with $a(y;\boldsymbol{\theta}) - \mathbb{E}[a(y;\boldsymbol{\theta})]$ given the fact that if $a(y;\boldsymbol{\theta})$ is continuous in $\boldsymbol{\theta}$ and satisfies the dominance condition i.e., bounded by integrable dominating function e.g., $\mathbb{E}[sup_{\boldsymbol{\theta}\in\Theta}|a(y;\boldsymbol{\theta})|] < \infty$, then $\mathbb{E}[a(y;\boldsymbol{\theta})]$ is continuous in $\boldsymbol{\theta}$.

$$
\begin{aligned}
lim_{\boldsymbol{\theta}\to\boldsymbol{\theta}'}\mathbb{E}[a(y;\boldsymbol{\theta})] &= \mathbb{E}[lim_{\boldsymbol{\theta}\to\boldsymbol{\theta}'}a(y;\boldsymbol{\theta})] && \text{by Dominated Convergence Theorem} \\
&= \mathbb{E}[a(y;\boldsymbol{\theta}')] && \text{by continuity of } a
\end{aligned}
$$

Remark 2: Lemma 2.4 of Newey & McFadden (1994) weakens the continuity assumption to continuity with probability one. The proof is neat and can be found in the appendix of Tauchen (1985).

Remark 3: The convergence in probability here can be strengthened to almost sure convergence with the same hypothesis (see Appendix).

1.4 Corollary 2 (Distribution function) (Example 19.6 of Van der Vaart (1998))

If $\mathcal{F} = \{\mathbb{1}(x \in (-\infty, t]), t \in \mathbb{R}\}$, Abstract Glivenko-Cantelli Theorem reduces to the (classical) Glivenko-Cantelli Theorem i.e., $||\mathbb{F}_n - F||_\infty := sup_{t\in\mathbb{R}}|\mathbb{F}_n(t) - F(t)| \rightarrow_{as} 0$.

Proof:

Given a $\varepsilon > 0$, there is a grid of points $-\infty = t_0 < t_1 < \cdots < t_k = \infty$ such that $F(t_i-) - F(t_{i-1}) < \varepsilon$ for $\forall i$ (only points at the partition can jump more than ε). We construct the bracket $[\mathbb{1}(x \in (-\infty, t_{i-1}]), \mathbb{1}(x \in (-\infty, t_i))]$,

$$\begin{aligned} ||\mathbb{1}(x \in (-\infty, t_i)) - \mathbb{1}(x \in (-\infty, t_{i-1}])||_{F,1} &:= F(\mathbb{1}(x \in (-\infty, t_i)) - \mathbb{1}(x \in (-\infty, t_{i-1}])) \\ &= F(t_i-) - F(t_{i-1}) < \varepsilon \end{aligned}$$

Therefore, the bracket $[\mathbb{1}(x \in (-\infty, t_{i-1}]), \mathbb{1}(x \in (-\infty, t_i))]$ has $L_1(F)$-size ε. Q.E.D.

1.5 Classical Glivenko-Cantelli Theorem (Direct Derivation, Theorem 19.1 of Van der Vaart (1998))

If $\{X_i\}_{i=1}^\infty$ is i.i.d with distribution function F. Then $||\mathbb{F}_n - F||_\infty := sup_{t\in\mathbb{R}}|\mathbb{F}_n(t) - F(t)| \rightarrow_{as} 0$.

Proof:

Given a $\varepsilon > 0$, there is a grid of points $-\infty = t_0 < t_1 < \cdots < t_k = \infty$ such that $F(t_i-) - F(t_{i-1}) < \varepsilon$ for $\forall i$ (only points at the partition can jump more than ε). For $t_{i-1} \le t < t_i$, note that $\mathbb{F}_n(t) \le \mathbb{F}_n(t_i-)$ and $F(t_i-) - F(t_{i-1}) < \varepsilon \iff F(t_i-) - F(t) < \varepsilon \iff -F(t) < \varepsilon - F(t_i-)$

$$\mathbb{F}_n(t) - F(t) \le \mathbb{F}_n(t_i-) - F(t) < \mathbb{F}_n(t_i-) - F(t_i-) + \varepsilon$$

Note also that $\mathbb{F}_n(t) \ge \mathbb{F}_n(t_{i-1})$ and $F(t_i-) - F(t_{i-1}) < \varepsilon \iff F(t) - F(t_{i-1}) < \varepsilon \iff F(t) < F(t_{i-1}) + \varepsilon$

$$\mathbb{F}_n(t) - F(t) \ge \mathbb{F}_n(t_{i-1}) - F(t) > \mathbb{F}_n(t_{i-1}) - F(t_{i-1}) - \varepsilon$$

For each $i \in \{1, \cdots k-1\}$, $\{\mathbb{F}_n(t_i-) - F(t_i-)\}_{n=1}^\infty$ and $\{\mathbb{F}_n(t_{i-1}) - F(t_{i-1})\}_{n=1}^\infty$ are sequences of i.i.d. random variables, both converge to zero almost surely by strong law of large numbers. Therefore,

$$|lim_{n\rightarrow\infty} sup_{t\in\mathbb{R}}(\mathbb{F}_n(t) - F(t))| = lim_{n\rightarrow\infty} sup_{t\in\mathbb{R}}|\mathbb{F}_n(t) - F(t)| := lim_{n\rightarrow\infty}||\mathbb{F}_n - F||_\infty < \varepsilon \qquad a.s.$$

Q.E.D.

1.6 Corollary 3 (Parametric Class, Example 19.7 of Van der Vaart (1998))

If $\mathcal{F} = \{f_\theta : \theta \in \Theta\}$ where (i) f_θ is a measurable function satisfying Lipschitz continuity i.e., there exists a measurable function m such that $|f_{\theta_1}(x) - f_{\theta_2}(x)| \le m(x)||\theta_1 - \theta_2||$ for every θ_1, θ_2. (ii) Θ is a bounded set and is a subset of $\mathbb{R}^d$. (iii)

$P|m|^r < \infty$. Then, there exists a $K(\Theta, d)$ such that $N_{[]}(2\varepsilon||m||_{P,r}, \mathcal{F}, L_r(P)) \leq K(diam(\Theta)/(2\varepsilon))^d$ for $0 < 2\varepsilon < diam(\Theta)$ (and $sup_{f\in\mathcal{F}}|\mathbb{P}_n f - Pf| \rightarrow_{as} 0$ by Abstract Glivenko-Cantelli Theorem)

Proof:
Given a $0 < \varepsilon < diam(\Theta)/2$, we pick an arbitrary θ_1 from Θ, for any θ_2 such that $||\theta_1 - \theta_2|| \leq \varepsilon$ i.e., from the closed ball with center θ_1 and radius ε, we have

$$|f_{\theta_1}(x) - f_{\theta_2}(x)| \leq m(x)||\theta_1 - \theta_2|| \leq m(x)\varepsilon$$

Thus,

$$\begin{aligned} -m(x)\varepsilon \leq f_{\theta_1}(x) - f_{\theta_2}(x) &\leq m(x)\varepsilon \\ -f_{\theta_1}(x) - m(x)\varepsilon \leq -f_{\theta_2}(x) &\leq -f_{\theta_1}(x) + m(x)\varepsilon \\ f_{\theta_1}(x) + m(x)\varepsilon \geq f_{\theta_2}(x) &\geq f_{\theta_1}(x) - m(x)\varepsilon \\ f_{\theta_1}(x) - m(x)\varepsilon \leq f_{\theta_2}(x) &\leq f_{\theta_1}(x) + m(x)\varepsilon \end{aligned}$$

Therefore, we construct the bracket $[f_\theta - \varepsilon m, f_\theta + \varepsilon m]$ with $L_r(P)$-size

$$\begin{aligned} ||f_\theta + \varepsilon m - (f_\theta - \varepsilon m)||_{P,r} &= ||2\varepsilon m||_{P,r} \\ &= (P|2\varepsilon m|^r)^{1/r} \\ &= (|2\varepsilon|^r P|m|^r)^{1/r} \\ &= 2\varepsilon||m||_{P,r} \end{aligned}$$

As the θ is arbitrary, we have $\{[f_\theta - \varepsilon m, f_\theta + \varepsilon m]\}_{\theta\in\Theta}$.

The largest size of Θ is $diam(\Theta) < \infty$ for any $d > 0$ (it is finite as Θ is bounded), the diameter of the closed ball defined above is 2ε. The largest number of closed balls needed to "cover" Θ is $(diam(\Theta)/(2\varepsilon))^d$ for any $d > 0$. Q.E.D.

Remark 1: Instead of assuming compact Θ in Corollary 1, Θ here is only bounded. However, the continuity assumption in Corollary 1 is strengthened to Lipschitz continuity. This example shows that the conditions of the typical uniform law of large numbers can be altered by using the Abstract Glivenko-Cantelli Theorem.

1.7 Uniform Law of Large Numbers with Bounded Parameter Space (Direct Derivation, Theorem 7.9 and 7.11 of Hansen (2020))

If $\{X_i\}_{i=1}^\infty$ is i.i.d. (i) $\mathbb{E}[|g(X_1;\boldsymbol{\theta})|] < \infty$ for $\forall\boldsymbol{\theta} \in \Theta$ (ii) Θ is a bounded set (iii) $g(x;\boldsymbol{\theta})$ is first-order Lipschitz. Then, $sup_{\boldsymbol{\theta}\in\Theta}|N^{-1}\sum_{i=1}^N g(X_i;\boldsymbol{\theta}) - \mathbb{E}[g(X_1;\boldsymbol{\theta})]| \rightarrow_p 0$.

Proof:
Combining Theorem 7.9 and 7.11 of Hansen (2020). The proof of Theorem 7.9 follows similar ideas of that of Theorem 4.2.1 of Amemiya (1985). Q.E.D.

Remark: (i) and (iii) of Corollary 3 implies that the function f_θ is first-order Lipschitz with $r = 1$. Corollary 3 does not require $\mathbb{E}[|g(X_1;\boldsymbol{\theta})|] < \infty$ (needed for the application of weak law of large numbers) and the convergence is stronger.

2 Reference

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3 Appendix

3.1 Proof of Strong Law of Large Numbers

3.1.1 Lemma 1

$$\sum_{i=1}^{\infty} Pr(|X| \geq i) \leq \mathbb{E}[|X|] \leq 1 + \sum_{i=1}^{\infty} Pr(|X| \geq i)$$

Proof: see Theorem 3.2.1 of Chung (2001). Q.E.D.

3.1.2 Borel-Cantelli Theorem

Let $\{A_i\}_{i=1}^{\infty}$ be a sequence of events on the probability space,
(a) If $\sum_{i=1}^{\infty} Pr(A_i) < \infty$, then $Pr(A_i \quad i.o.) = 0$;
(b) If A_i is independent and $\sum_{i=1}^{\infty} Pr(A_i) = \infty$, then $Pr(A_i \quad i.o.) = 1$.

Proof: see Rao (1973, p.137-138). Q.E.D.

3.1.3 Kolmogorov Theorem 1

Let $\{X_i\}_{i=1}^{\infty}$ be a sequence of independent (not necessarily identically distributed) random variables such that $\mu_i := \mathbb{E}[X_i]$ and $\sigma_i^2 := Var(X_i)$, we have

$$\sum_{i=1}^{\infty} \frac{\sigma_i^2}{i^2} < \infty \implies N^{-1} \sum_{i=1}^{N} (X_i - \mu_i) \to_{as} 0$$

Proof: see Rao (1973, p.114). It is a consequence of Hajek-Renyi inequality. Q.E.D.

3.1.4 Kolmogorov Theorem 2

This theorem is frequently used in the proofs of this note so we prove it here for the sake of completeness.

Let $\{X_i\}_{i=1}^{\infty}$ be a sequence of i.i.d. random variables, we have

$$\mu := \mathbb{E}[X_1] < \infty \iff N^{-1} \sum_{i=1}^{N} X_i \to_{as} \mu$$

Proof:
($\Longleftarrow$) (This part of the proof is from Rao (1973, p.115))

$$\begin{aligned} \frac{X_N}{N} &= N^{-1} \sum_{i=1}^{N} X_i - \frac{N-1}{N} (N-1)^{-1} \sum_{i=1}^{N-1} X_i \\ &\to_{as} \mu - 1 \cdot \mu = 0 \qquad\qquad \text{as } N^{-1} \textstyle\sum_{i=1}^{N} X_i \to_{as} \mu \end{aligned}$$

This implies that $Pr(|\frac{X_N}{N}| \geq 1 \quad i.o.) = Pr(|X_N| \geq N \quad i.o.) = 0$ where $i.o.$ means "infinitely often". Moreover, the independence of X_i implies the independence of the event $|X_N| \geq N$. Therefore, by Borel-Cantelli Theorem (using the logic If not B then not A for (b)), we have

$$\sum_{N=1}^{\infty} Pr(|X_N| \geq N) < \infty$$

By Lemma 1 above, we have $\mathbb{E}[|X_N|] < \infty$. This implies that $\mu := \mathbb{E}[X_1] < \infty$.

($\implies$) (This part of the proof is from Theorem 5.4.2 of Chung (2001))
Define

$$Y_i := \begin{cases} X_i & \text{if } |X_i| \leq i \\ 0 & \text{if } |X_i| > i \end{cases}$$

We apply $\{Y_i - \mathbb{E}[Y_i]\}_{i=1}^{\infty}$ to Kolmogorov Theorem 1 (note that it is not identically distributed and $Var(Y_i - \mathbb{E}[Y_i]) = Var(Y_i)$).

$$\begin{aligned}
\sum_{i=1}^{\infty} \frac{Var(Y_i)}{i^2} &= \sum_{i=1}^{\infty} \frac{\mathbb{E}[Y_i^2] - \mathbb{E}[Y_i]^2}{i^2} \\
&\leq \sum_{i=1}^{\infty} \frac{\mathbb{E}[Y_i^2]}{i^2} \\
&= \sum_{i=1}^{\infty} \frac{1}{i^2} \int_{-\infty}^{\infty} y^2 dF_Y(y) \\
&= \sum_{i=1}^{\infty} \frac{1}{i^2} \int_{|y| \leq i} y^2 dF_Y(y) \\
&= \sum_{i=1}^{\infty} \frac{1}{i^2} \int_{|x| \leq i} x^2 dF_X(x) && \text{as } Y_i = X_i \text{ if } |Y_i| = |X_i| \leq i \\
&= \sum_{i=1}^{\infty} \frac{1}{i^2} \sum_{j=1}^{i} \int_{j-1 < |x| \leq j} x^2 dF_X(x) \\
&= \sum_{j=1}^{\infty} \int_{j-1 < |x| \leq j} x^2 dF_X(x) \sum_{i=j}^{\infty} \frac{1}{i^2} && \text{write down the elements and see the pattern} \\
&= \sum_{j=1}^{\infty} \int_{j-1 < |x| \leq j} |x|^2 dF_X(x) \sum_{i=j}^{\infty} \frac{1}{i^2} \\
&\leq \sum_{j=1}^{\infty} \int_{j-1 < |x| \leq j} j \cdot |x| dF_X(x) \sum_{i=j}^{\infty} \frac{1}{i^2} \\
&\leq \sum_{j=1}^{\infty} j \int_{j-1 < |x| \leq j} |x| dF_X(x) \frac{C}{j} && \textstyle\sum_{i=j}^{\infty} \frac{1}{i^2} \leq \frac{C}{j} \text{ for some } C \\
&= C \sum_{j=1}^{\infty} \int_{j-1 < |x| \leq j} |x| dF_X(x) \\
&\leq C \sum_{j=1}^{\infty} \int_{j-1 \leq |x| \leq j} |x| dF_X(x) \\
&= C \int_{0 \leq |x| < \infty} |x| dF_X(x) \\
&= C\mathbb{E}[|X_1|] < \infty && \text{implies by } \mathbb{E}[X_1] < \infty
\end{aligned}$$

Therefore, we have $N^{-1} \sum_{i=1}^{N} (Y_i - \mathbb{E}[Y_i]) \to_{as} 0$ (note that $\mu_i = \mathbb{E}[Y_i - \mathbb{E}[Y_i]] = 0$) by Kolmogorov Theorem 1.

Note that $\mathbb{E}[Y_i] = \int_{-\infty}^{\infty} y dF_Y(y) = \int_{|y| \leq i} y dF_Y(y) = \int_{0 \leq |x| \leq i} x dF_X(x)$. Thus, $lim_{i \to \infty} \mathbb{E}[Y_i] = lim_{i \to \infty} \int_{0 \leq |x| \leq i} x dF_X(x) = \mathbb{E}[X_1]$. Therefore,

$$N^{-1} \sum_{i=1}^{N} (Y_i - \mathbb{E}[Y_i]) \to_{as} 0$$

$$N^{-1} \sum_{i=1}^{N} Y_i \to_{as} N^{-1} \sum_{i=1}^{N} \mathbb{E}[Y_i] \to \frac{N}{N} \mathbb{E}[X_1] = \mathbb{E}[X_1]$$

as $N \to \infty$. Now we show that $\{X_i\}_{i=1}^{\infty}$ and $\{Y_i\}_{i=1}^{\infty}$ are equivalent,

$$\sum_{i=1}^{\infty} Pr(X_i \neq Y_i) = \sum_{i=1}^{\infty} Pr(|X_i| > i) = \sum_{i=1}^{\infty} Pr(|X_1| > i) \leq \mathbb{E}[|X_1|] < \infty$$

by Lemma 1 and hypothesis $\mathbb{E}[X_1] < \infty$. Thus, $N^{-1}\sum_{i=1}^{N} X_i \to_{as} \mathbb{E}[X_1]$ Q.E.D.

3.2 Proof of Weak Law of Large Numbers

3.2.1 Levy's Continuity Theorem

Let $\phi_j(t)$ be the characteristic function of X_j,

(a) $X_j \to_d X \implies \phi_j(t) \to \phi(t)$ pointwise in t where $\phi(t)$ is the characteristic function of X;

(b) $\phi_j(t) \to \phi(t)$ pointwise in t and $\phi(0)$ is continuous $\implies \phi(t)$ is a characteristic function of a random variable, say X, and $X_j \to_d X$.

Proof:

(a) can be proved by a direct application of Portmanteau Lemma (Lemma 2.2 (ii) of Van der Vaart (1998)) as $exp(itz)$ is a bounded and continuous function of z. See Rao (1973, p.119) for the proof of (b) which applies Helly Lemma. Q.E.D.

Remark: This theorem is also a key ingredient in the proof of Lindeberg-Levy Central Limit Theorem.

3.2.2 Khinchine's Theorem

Let $\{X_i\}_{i=1}^{\infty}$ be a sequence of i.i.d. random variables, we have

$$\mu := \mathbb{E}[X_1] < \infty \implies N^{-1}\sum_{i=1}^{N} X_i \to_p \mu$$

Proof:

(This is from the Proposition 2.16 of Van der Vaart (1998)) Let $\phi(t)$ be the characteristic function of X_i. The first-order Taylor expansion of $\phi(t)$ centered at 0 is

$$\begin{aligned}\phi(t) &= \phi(0) + \phi'(0)t + o(t) \\ &= 1 + i\mu t + o(t) && \mu < \infty \implies \phi'(0) = i\mu\end{aligned}$$

Therefore,

$$\begin{aligned}\mathbb{E}_X[exp(itN^{-1}\sum_{j=1}^{N} X_j)] &= \mathbb{E}_X[\prod_{j=1}^{N} exp(i\frac{t}{N}X_j)] \\ &= \prod_{j=1}^{N} \mathbb{E}_X[exp(i\frac{t}{N}X_j)] && \text{by independence of } X_i \text{ and Fubini's Theorem} \\ &= (\phi(t/N))^N && \text{by identical distribution of } X_i \\ &= (1 + i\mu\frac{t}{N} + o(\frac{t}{N}))^N && \text{by Taylor expansion above} \\ &\to (exp(i\mu\frac{t}{N}))^N && \text{using } ln(1+x) \approx x \text{ by Taylor expansion} \\ &= exp(it\mu) \\ &= \mathbb{E}_\mu[exp(it\mu)] && \text{as } \mu \text{ is a constant}\end{aligned}$$

Note that $\mathbb{E}_\mu[exp(it\mu)]$ is continuous at $t = 0$. Thus, $N^{-1}\sum_{j=1}^{N} X_j \rightarrow_d \mu$ by Levy's Continuity Theorem. As $X \rightarrow_p Y \iff X \rightarrow_d Y$ if Y is a constant, we have $N^{-1}\sum_{j=1}^{N} X_j \rightarrow_p \mu$ Q.E.D.

Remark 1: This can also be proved without using Levy's Continuity Theorem e.g., see Rao (1973, p.112-113)

Remark 2: With the same hypothesis of Strong Law of Large Numbers i.e., i.i.d. $\{X_j\}_j$ and finite $\mathbb{E}[X_1]$, Weak Law of Large Numbers says that the sample average only converges to $\mathbb{E}[X_1]$ in probability, instead of almost surely. In general, $X \rightarrow_{as} Y \implies X \rightarrow_p Y$. So, the Weak Law says something that is weaker. The proof of it is less involved as a result.

Remark 3: Sample average converging in probability to the first moment of X_j is implied by Chebyshev's inequality. However, it requires a finite variance of X_j (but does not require the independence of $\{X_j\}_j$).

3.2.3 Chebyshev's Inequality

For any random variable X with $\mu := \mathbb{E}[X]$ and $\sigma^2 := Var(X) < \infty$. For any $\lambda > 0$,

$$Pr(|X - \mu| \geq \lambda\sigma) \leq \frac{1}{\lambda^2}$$

If we set $\lambda = \varepsilon/\sigma$, we have

$$Pr(|X - \mu| \geq \varepsilon) \leq \frac{\sigma^2}{\varepsilon}$$

Proof:

(This neat proof is from Rao (1973, p.95))

$$\begin{aligned}
\sigma^2 := Var(X) &= \int_{-\infty}^{\infty} (x-\mu)^2 dF_X(x) \\
&= \int_{-\infty}^{\mu-\lambda\sigma} (x-\mu)^2 dF_X(x) + \int_{\mu-\lambda\sigma}^{\mu+\lambda\sigma} (x-\mu)^2 dF_X(x) + \int_{\mu+\lambda\sigma}^{\infty} (x-\mu)^2 dF_X(x) \\
&\geq \int_{-\infty}^{\mu-\lambda\sigma} (x-\mu)^2 dF_X(x) + \int_{\mu+\lambda\sigma}^{\infty} (x-\mu)^2 dF_X(x) \\
&\geq \int_{-\infty}^{\mu-\lambda\sigma} (\mu-\lambda\sigma-\mu)^2 dF_X(x) + \int_{\mu+\lambda\sigma}^{\infty} (\mu+\lambda\sigma-\mu)^2 dF_X(x) \quad \text{the inequality holds because of the squares} \\
&= \lambda^2\sigma^2 \int_{-\infty}^{\mu-\lambda\sigma} dF_X(x) + \lambda^2\sigma^2 \int_{\mu+\lambda\sigma}^{\infty} dF_X(x) \\
&= \lambda^2\sigma^2 (\int_{-\infty}^{\mu-\lambda\sigma} dF_X(x) + \int_{\mu+\lambda\sigma}^{\infty} dF_X(x)) \\
&= \lambda^2\sigma^2 Pr(|X-\mu| \geq \lambda\sigma)
\end{aligned}$$

Q.E.D.

Remark 1: This holds for any random variables drawn from any sampling schemes e.g., i.i.d., i.n.i.d., dependence, etc.

Remark 2: If μ is not $\mathbb{E}[X]$, the inequality still holds if σ^2 is defined as $\mathbb{E}[(X-\mu)^2]$ (not a variance as μ is not a mean).

3.2.4 Chebyshev's Theorem

For any sequence of random variables $\{X_i\}_{i=1}^{\infty}$ with $\mu_i := \mathbb{E}[X_i]$ and $\sigma_i^2 := Var(X_i) < \infty$ and $Cov(X_i, X_j) = 0$ for $i \neq j$, we have

$$\frac{\sum_{i=1}^{N} \sigma_i^2}{N^2} \rightarrow 0 \implies N^{-1}\sum_{i=1}^{N} X_i \rightarrow_p N^{-1}\sum_{i=1}^{N} \mu_i$$

as $N \to \infty$

Proof:
First, note that $Var(N^{-1}\sum_{i=1}^{N} X_i) = N^{-2}Var(\sum_{i=1}^{N} X_i) = N^{-2}\sum_{i=1}^{N}\sigma_i^2$ because the covariances are zero by the hypothesis.

By Chebyshev's Inequality and the hypothesis, we have for any $\varepsilon > 0$,

$$Pr(|N^{-1}\sum_{i=1}^{N} X_i - N^{-1}\sum_{i=1}^{N}\mu_i| \geq \varepsilon) \leq \frac{\sum_{i=1}^{N}\sigma_i^2}{N^2\varepsilon} \to 0$$

as $N \to \infty$ Q.E.D.

Remark 1: This holds for any sequence of random variables drawn from any sampling schemes e.g., i.i.d., i.n.i.d., dependence, etc. If the identical distribution is assumed, $N^{-1}\sum_{i=1}^{N} X_i \to_p \mu$ as $\sigma^2/N \to 0$ and $\sigma^2 < \infty$ as $N \to \infty$. Compared with Khinchine's Theorem, it additionally assumes a finite second moment, but it does not assume the independence.